

Program overview

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Year 2023/2024
Organization Industrial Design Engineering
Education Master Integrated Product Design

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Master Industrial Product Design							
1st year Master Industrial Product Design							
General Courses Master Industrial Product Design							
ID4010	Design Theory and Methodology	3					
ID4060	Manage your Master	2					
ID4070	IDE Academy	4					
Master Specific Courses Industrial Product Design							
ID4120-23	Technical Ergonomics and Product Experience	3					
ID4130-23	Knowledge and Intelligence Design	3					
ID4135-23	Modelling	5					
ID4140-23	Material-Driven Design	5					
ID4170-23	Advanced Concept Design	18					
ID4175-23	Advanced Embodiment Design Project	10					
ID4180	Managing Product Innovation	3					
ID4185-23	Strategic and Sustainable Design	3					
2nd year Master Industrial Product Design							
ID4190-16	Graduation Project (IPD)	30					
Electives (min. 30 EC) IPD							
Specialization Medisign							
Compulsory Electives (min. 9 ECTS) Medisign							
Compulsory Electives (min. 9 EC), at least 2 EC IDE Electives							
ID5301	Biomechanics	3					
ID5383	Inclusive Design	3					
ID5663	Contextmapping Skills	6					
ID5721	Capita Selecta Medisign	3					
ID5732	Health Psychology: Tools and Methods	3					
ID5741-23	eHealth Design	3					
ID5752	Rules and Regulations for Designing Medical Devices	3					

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Master Industrial Product Design

Introduction 4

A mandatory application for all courses (mandatory and electives courses) shall be made via the electronic applications system Osiris in the period that Osiris there to is opened. This period closes about 5 calendar weeks prior to the first day of the semester in which the programme starts; the faculty announces the exact deadline for application timely. The application for courses has to take place per semester, meaning for courses of 2 quarters at once.

See <http://www.io.tudelft.nl/osiris> for deadlines and details.

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1st year Master Industrial Product Design
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General Courses Master Industrial Product Design

ID4010	Design Theory and Methodology	3
Course Coordinator	Dr. R.S.K. Chandrasegaran	
Contact Hours / Week x/x/x/x	3/0/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	The aim of this course is to help you gain a deeper understanding of the principles, practices and procedures of designing in a broad and general sense. Unlike other Universities, Delft has a rich research and education tradition in design theory and methodology. You have almost certainly heard of design thinking, for example, but did you know that Delft played a key role in its development? Through the course we will cover many topics and approaches but our core principle is to link different types of theory to design practice in a way that is insightful, useful, and entertaining. The course is NOT about learning new methods of designing, it is about reflecting on the concepts, practices, activities, and methods that designing is constituted by. Design Theory and Methodology (DTM) will help you to build a more nuanced vocabulary when talking about the design process thus aiding your development as a designer. Through a number of lectures, critical discussions, and practical activities you will improve your ability to describe, analyse, and understand the activity of designing. This will help you to reflect further on your own design processes, and those of others. Developing an ability to critically analyse design processes will empower you to change, modify, and adapt your thinking and acting while designing.	
Study Goals	By the end of this course it is expected that you will be able to: Learning Objective 1: Articulate theoretical aspects of design practice Learning Objective 2: Critically question design theories, methodologies, and practices Learning Objective 3: Demonstrate a theoretical understanding of your own design practice	
Education Method	The course is centred around a series of lectures and practicums that explore aspects of designing and the design process. A number of activities will help deepen your understanding and coaches will provide support for you at appropriate points. Online resources will help guide you through the course and include: MS Teams discussion forums, podcasts, key references and other relevant materials.	
Literature and Study Materials	The course will be managed on Brightspace with timetable, assignments, and course materials - including podcasts - available. Microsoft Teams will be used to manage discussion and share information during the course.	
Assessment	Assignment 1 The first two weeks of the course are about sensitising you to the concepts of design theory and methodology. Assignment 1 is a multiple-choice quiz to test your comprehension. Assignment 2 Following Assignment 1 you will participate in three practical sessions, which will involve small group work. A reflection on what you did in these sessions, and what you learned, make up your submission for the second assignment. Assignment 3 The final assignment is an individual 1500 word essay and short reflection. The essay is your chance to explore one specific topic area from the course. You will choose an essay topic from a number of options and develop an argument around a theme drawing on references from design research journals. Scholarly work forms the core of the assignment.	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Course Coordinators 2022-2023: Senthil Chandrasegaran, Assistant Professor Peter Lloyd, Professor Email: DTM-IO@tudelft.nl	

ID4060	Manage your Master	2
Course Coordinator	Ir. S.H. Gieles	
Contact Hours / Week x/x/x/x		
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	MYM introduces students to IDE Community and the structure and possibilities of the MSc program at IDE. It supports students in developing their own, personal vision on their Master education and in developing a plan for the final year of their Master program. MYM takes place on the first three days of the 1st and 2nd semester of their Master program. Each one of these days has a specific theme: Monday in first semester: Welcome to IDE Tuesday in first semester: Your & Our Identity Wednesday in first semester: Opportunities at IDE Monday in second semester: Your 3rd semester Tuesday in first semester: Research Wednesday in second semester: Preparing for graduation Every day provides a combination of activities to build community and help shape your master.	
Study Goals	After following this course, the student is able to 1. Describe what it means for them to be part of IDE Community 2. Formulate their own, personal vision on being a designer 3. Evaluate what they would like to develop further in the field of Industrial Design Engineering 4. Evaluate what are their personal interest areas in the field of Industrial Design Engineering 5. Formulate a plan for developing and filling in their 2nd Master year	
Education Method	The MYM days consist of lectures, group activities, workshops, individual assignments and presentations.	
Literature and Study Materials	Will be provided during the course	
Assessment	During the MYM course you prepare a personal development vision, including a plan for your 3rd and 4th semester. The idea of the vision is that it is a personal presentation and discussion document, in which you present a profile of yourself as Master student (DfI, SPD or IPD), you indicate your designer identity and development ambition, and what you plan to do in order to get there. This deliverable should be uploaded on Brightspace. In the 5 page master plan you display: (1) what characterizes you as a student of your particular master? why did you choose this master? how do you plan to fill it in, in your unique way? (LO1, LO2) (2) what are your current skills? what are you good at? what skills would you like to develop? consider different types of skills: technical, design, research, organizational and people skills. (LO3) (3) what are your dreams in your professional life? what are your passions? how do you see your future? in what kind of organisation would you like to work (profit/non-profit, own company, big/small)? where do you plan to work? how do you prepare for this career? (LO4) After completing the MYM days in the 1st semester, you need to upload the first three pages on Brightspace. (4) the courses you plan to follow in the 3rd semester: electives, Design-XL, internship, research project, external courses, and so on. Also think about how you prepare for these courses. (LO5) (5) your desired topic of your graduation project in the 4th semester, including suggestions for supervisors and involvement of external organisations (e.g., companies); also think about a back-up plan, in case things work out differently (LO5) After completing the MYM days in your 2nd semester, you need to upload all five pages in Brightspace. This time, your MYM plan will be assessed in terms of Pass or Fail. You can submit your master plan twice a year on Brightspace. Submission deadlines are at the end of the third week in Q1 and Q2.	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Remarks	Students should register for the MYM course at the start of the first semester of their Master program. You only need to register once for this course; you do not need to register again in your 2nd or 3rd semester.	

ID4070	IDE Academy	4
Course Coordinator	Ir. A.J.M. van Leeuwen	
Contact Hours / Week x/x/x/x		
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The IDE Academy is a series of one-day workshops, always on Wednesdays, on design-related skills, knowledge and insights. Each Master's student enrolls in the academy to participate in 14 such workshops (7 workshops for exchange students), rounding off enrollment with a personal reflection. A typical academy workshop provides basic training on a relevant set of transferrable IDE skills. Sample topics relate to research (e.g. observational research, literature surveying), applying methods (e.g. scrum, business models, innovation management), conceptual design (e.g. visualisation skills, generative design), prototyping (e.g. screen-based prototypes, programming), technology (e.g. VR tools, smart materials), and personal skills (e.g. team work, reporting). In most cases, the workshop also provides an entry point for further self-study and development. Industry events, such as business/technology fairs, can also double as workshops: the student actively participates in the event and are given an additional assignment to reports back at the end of the day. The IDE academy makes good use of different networks (alumni, research and practice) by inviting a wide range of design practitioners from industry. This way, IDE Academy builds a bridge between academy and industry.	
Study Goals	- Exploration of diverse skills (14 workshops, 7 for Exchange students) - Recognition of skill improvement - Recognition of the value of skills for their career - Recognition of transferability of the skills learned to other courses	
Education Method	As IDE Academy staff, we encourage students to take responsibility for their own learning, and motivate them to independently seek for new skill development, knowledge, and design insights. We do this by offering around 100 workshops a year over almost 35 weeks. Students attend 14 workshops (7 for exchange students) of their choice. Students can start with The IDE Academy at the beginning of every semester that always kicks off with a workshop where new students can get familiarised with the IDE Academy. In this introductory workshop for both new and current students the workshops for the upcoming semester are presented and they are asked to make a plan for their learning. Every week, students can register (enroll) for a workshop of their choice. Students do not have to do all workshops consecutively. That is not possible anyway because the number of places offered is usually not enough for everyone to participate. They can plan the workshop as they see fit. If enrollment exceeds the maximum of participants, places are randomly assigned. There is a system that guarantees places if students have been rejected too many times. Students are encouraged to complete all workshops of the IDE Academy in one year In most workshops, students actively engage with the subject matter in small groups. This allows for a variety of group work themes to develop. During the workshop, the student's attendance is registered. Immediately after completion, students provide brief feedback about the workshop by means of a quiz and rank it on a 5-point scale from "not useful" to "highly useful". This feedback is then used by the staff to improve the portfolio of workshops and make improvements. Registration of attendance and the quiz are both compulsory to validly complete a workshop. After completing the workshops students are asked to write a personal reflection on their learning by means of a "Letter to Future You". Due to the Master renewal that will be introduced next academic year, IDE Academy students are encouraged to complete the course at the end of this academic year (2023/2024).	
Assessment	Assessment is based on attendance which has to be registered on the Attendance List during the workshop AND by completing the quiz at the end of every workshop. At any time after successfully completing the 14 workshops (7 for exchange students), students have to reflect on the course and the workshops by writing a letter to their future self.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

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Master Specific Courses Industrial Product Design
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ID4120-23	Technical Ergonomics and Product Experience	3
Course Coordinator	Dr. rer. nat. C. Schneegass	
Contact Hours / Week x/x/x/x		
Education Period	3	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	The Technical Ergonomics and Product Experience course focuses on the theory and practice of designing, evaluating and optimising human product experience. It provides an introduction into fundamental theories and models in cognitive and physical ergonomics (e.g., mental processing, comfort, etc.), and teaches students how to plan and conduct empirical research for measuring and evaluating product experiences.	
Study Goals	Upon successful completion of the course students will: 1. Explain relevant principles of human cognitive and physical ergonomics and their influence on product experience. 2. Compare approaches and methods for evaluating cognitive and physical ergonomics with regard to their suitability in a given research setup. 3. Analyse the product experience of a given product or system along relevant theoretical frameworks. 4. Justify design decisions based on theoretical and empirical insights.	
Education Method	The course provides scientific foundations of physical and cognitive ergonomics, and theories, analytical frameworks and methods for measuring product experiences. This theoretical part will be taught through a combination of lectures, assigned readings and self-study. Students will be assessed individually through a written exam. A formative quiz allows students to test their understanding and prepare for the final exam. The course closely follows the Advanced Embodiment Design Project and asks students to apply their theoretical understanding to inform and justify their product design with respect to physical and cognitive ergonomics criteria. The applied part will be assessed through a group assignment with a written deliverable that synthesises product experience and ergonomics knowledge.	
Assessment	The course will be assessed by: Individual written exam (50%) Group assignment (50%)	
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ID4130-23	Knowledge and Intelligence Design	3
Course Coordinator	Prof. G.W. Kortuem	
Contact Hours / Week x/x/x/x		
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	This course provides an in-depth understanding of intelligent technologies and intelligent products & services. It examines research findings and recent thinking about advanced digital technologies (such as Internet of Things, Artificial Intelligence, machine learning, robotics, and data analytics) and critically explores the nature and implications of machine intelligence. It examines contributions from design, human-computer-interaction and engineering, and gives students the opportunity to consider issues in designing effective and appropriate intelligent products & services. This course builds on knowledge and skills taught in various IDE-BSc courses, such as Digital Product Development, Digital Interfaces, and Data, and acts as a guide to digital technology master electives such as ID5515 Advanced Machine Learning for Design, ID5482 Data processing and Analytics, ID5417 Artificial Intelligence and Society, ID5423 Urban Analytics and ID5514 Crowd Computing for Design.	
Study Goals	After completing the course students will be able to: - explain capabilities of advanced intelligent technologies - analyze, evaluate and critique technical, human and ethical aspects of intelligent products/services	
Education Method	This course uses a case study approach rooted in concrete examples of intelligent products and services. Students will investigate existing products/services or research prototypes through lectures, classroom discussions and required readings. Students will advance their understanding through exercises focused on specific technology and design aspects.	
Assessment	Assessment is 100% individual based on a final examination (50%) and course work (50%).	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

ID4135-23	Modelling	5
Course Coordinator	Dr. J.H. Boyle	
Contact Hours / Week x/x/x/x		
Education Period	3	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Modelling is a valuable tool for helping us to understand the world we live in and predict what will happen under different circumstances. Effects of products and systems in the physical world, like overheating and mechanical failure, have a cause. These effects are predicable, at least in principle, if we understand the rules of the system well enough. The goal of modelling is to identify, quantify, and apply the rules underlying the behaviour of a physical system. In the context of product design, models can help us predict how the system we are designing (or its components) will perform in certain scenarios and can help us optimise the performance of our designs. This course builds on or complements the knowledge taught in several courses in the IDE-BSc programme (e.g Product Engineering, Design Analytics, Machine Learning for Design) and addresses modelling techniques in greater breadth, depth and technical detail. The course focusses primarily on four domains of physical modelling: solid mechanics, dynamical systems, fluid mechanics and thermodynamics. It teaches you about the underlying theory and both analytical and computational approaches to modelling these phenomena. It also addresses the necessary underpinning knowledge including representation of position, motion and shape, and a bit of relevant mathematics. Throughout the course, you will learn about the benefits, limitations and risks of modelling, and how to apply modelling techniques critically and appropriately.	
Study Goals	Upon successful completion of the course, students will be able: 1. to describe and explain the theoretical foundations of modelling in the domains of: solid mechanics, dynamic systems, fluid mechanics and thermodynamics. 2. to apply theoretical and computational models in the domains of: solid mechanics, dynamic systems, fluid mechanics and thermodynamics to the analysis of a system or system component. 3. to synthesize the results or predictions from multiple modelling studies in order to draw conclusions about or make recommendations for the overall functionality of a system or system component. 4. to evaluate the suitability of a particular model for representing a system or system component, and the validity or accuracy of that model's results.	
Education Method	Each week of the course is devoted to a different modelling topic, beginning with broadly applicable fundamentals and moving on to specific modelling domains. The weekly topics are taught and reinforced through: Lectures Practical workshops Assigned reading In addition, these distinct topics are combined and applied in a product design context through: Work on the group assignment	
Assessment	The course will be assessed by: One written exam (60%) One group assignment (40%)	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

ID4140-23	Material-Driven Design	5
Course Coordinator	Dr. E. Karana	
Contact Hours / Week x/x/x/x		
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Material Driven Design (MDD) Course is a didactically attractive program with a strong focus on designing with emerging materials and new fabrication technologies in the interest of the planet and society. Conventionally in product design, designers material considerations come at the very end of the design process where materials are selected from a pool of candidate materials to fulfill desired product function. While this approach still concerns majority of design projects in industry, over the last decade professionals in design have been increasingly involved in the development of new materials and products in synergy. MDD course advocates that designers have a unique position in unlocking potentials of emerging materials for novel functions, user experiences, and sustainable solutions in design. This requires a fundamental shift in the way we understand materials in design- as given and fixed to approaching materials generatively, for all they have to offer. In this course, students will practice the Material Driven-Design Method, which motivates a return to making and material tinkering to discover novel affordances of materials and its process-ability. It also suggests techniques to understand and integrate the technical properties of a material (function), its experiential qualities (i.e., how it is received by end users) (experience), and its positive contribution for the planet (sustainability) in unveiling the design potential of the material. The MDD method allows a back and forth thinking between details, i.e., material studies focusing on material characterisation, and wholeness, i.e., the way in which the material is approached from the perspective of user, appraised within a product, and discussed for the general good.	
Study Goals	Students who follow the course should: 1. be able to apply and substantiate a design method to tackle design projects in which a particular material is the departure point; 2. be able to select and substantiate techniques and tools to bridge technical and experiential considerations in material-driven design; 3. be able to characterize materials technically and experientially; 4. be able to zoom out and reflect on the wider implications of the material for sustainability; 5. be able to create material concepts through systematic tinkering; 6. be able to transfer their findings into a vision for a material application.	
Education Method	Students will run group projects (as groups of 5) in which a specific material will be a departure point in the design process. We will run 5-6 material stations in parallel (e.g., Meta-Materials Station, Living Materials Station, Animated Textiles Station), each focusing on an emerging material which our IDE faculty has expertise in. These projects will give particular attention to emerging materials with unique temporal qualities (i.e., the materials that change and evolve with external stimuli) to train our students with a competency to tackle the complex challenges of adaptive, autonomous, and living materials in our future world. Each station will be led by a material expert from IDE, and will have max 30 students. Students will be asked to select a project they desire to focus on through the course. These projects will provide experimental underpinning of the MDD techniques, which will be conveyed through lectures, workshops, and supported by tools (e.g., material taxonomy template, experiential characterization toolkit). The theoretical content of the course will be organised under 5 modules: [1] Creative Making (how to open up the design space in material tinkering) [2] Characterising Temporality (how to capture, analyse, and anticipate change in materials) [3] Understanding Material-People Relations (how to characterise materials experientially) [4] Vision in Material Driven Design (how to reflect on the purpose of the material; how to integrate sustainable thinking in material-driven design) [5] From Material to Product (how to come up with unique material applications that make sense)	
Assessment	The course will end with an exhibition in which students will present their ideas and their material driven design process to their peers, the course coordinator, and the material station experts, who will also give feedback on the final outcome. Prior to the exhibition, students will be asked to deliver an A4, including a title, a short abstract (max 300 words), and a graphical abstract of their project (template will be provided). They will be asked to give a scientific conference presentation at the final event, covering the following aspects as main deliverables: MDD Report (Group of 4/5 students) (70% of the final grade), Final Exhibition/Presentation (Group of 4/5 students) (30% of the final grade).	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

ID4170-23	Advanced Concept Design	18
Course Coordinator	Dr.ir. M.C. Dekker	
Contact Hours / Week x/x/x/x		
Education Period	1 2	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	The course Advanced Concept Design (ACD) focuses on the conceptualization phase of product development and starts from a design challenge and ends with a concept of a product or a product-service system. The initial real-life design challenge is co-created by the ACD coordination team and a partner organization (a company mostly a multinational, an IDE research group, or a semi-governmental body like a hospital or an airport). The final concept is elaborated in a sense of cultural & societal impact, usability, aesthetics and (emerging) technologies. The course is built around a Concept Design Project in combination with three areas of expertise, and has a strong focus on research. These areas of expertise are: Culture Ergonomics Visualization In these areas, students acquire specialized theoretical knowledge and skills through lectures, assignments and workshops. Scheduled feedback moments include coach and expert meetings and presentations, to assess the competencies of the students in these areas and their application in the Concept Design Project. Concept Design Project (Concept Design 1 and Concept Design 2) In the first period (Concept Design 1), a design team existing of about 5 students starts with deduction of the design challenge, exploring the client organization and performing design related research within the scope of the design assignment concerning the different areas of expertise. Furthermore, students explore, select and analyse, existing and emerging technologies inspirational for their project. They search for the so-called innovative concept drivers and therefore have to weigh and prioritize the collected information about cultural & societal impact, user needs and potential technologies. At 2/3 of this period, students define their individual design vision - what to design, for whom, and for which future scenario - based on the previous research outcomes and client analysis. This period finishes with a presentation on the group research results and individual vision and ideations. In the second period (Concept Design 2), students work individually on developing their ideas into a feasible concept. The conceptualization process encourages students to strive for societal impact, innovative usage, feasibility in terms of applied technologies, and estimated economic benefit. At several moments during the Concept Design Project, domain related simulation techniques are implemented in the project to evaluate and elaborate preliminary and final design concepts. Students advise their client on the required steps to take to implement their future innovation. This period finishes with a movie presentation and report on the final concept and contributing cultural, ergonomic and technological studies. Culture In this area of expertise, theory and methods are offered concerning the role designed artefacts and services play in shaping society. The Culture area focuses on cultural studies (related to e.g. material culture, design history and cultural anthropology) and philosophy of technology. Insight is gained in the impact of products on behavior and society, and the aligning of products to cultural and social practices and processes. Furthermore, awareness is created about the importance of products design concepts and the impact of their visual appearance. How does the concept influence the chosen context? What societal values and needs are addressed with the designed concept? Ergonomics In this area of expertise, research methods as supportive tools in ergonomic product design are presented in lectures, covering the gathering of user-, usage- and contextual data and their analysis. Academic experts and guest speakers pass on this theoretical and practice based knowledge to the students. Students have to prove their ability to select and apply these methods like user observation and interviews, for the benefit of their Concept Design Project. They are also provided with knowledge about setting up and performing concept evaluations. User insights and preferences in terms of (dis)comfort, safety and usability are identified and provide inspiration and knowledge for the design decisions throughout the Concept Design Project. Visualization Visual communication means understanding visual language and having knowledge of visual perception and communication. In Visualization, professional communication and presentation skills are introduced as an essential part of the daily practice of product design. As presentation techniques and portfolio work become increasingly important to designers who want to stand out and stay competitive, students acquire this understanding of visual information. This is offered not only through a theoretical approach, but also by working on practical exercises aimed at an understanding of the theory. At the same time, students improve their drawing skills through the use of new media. With topics derived from daily practice in design studios, students perform exercises in quick sketching techniques concerning: communication, problem-solving, conceptualization, presentation and portfolio design work. Finally, making a personal portfolio implies the use of visualization software (e.g. Photoshop, Illustrator and Indesign) and digital photography.	
Course Contents Continuation		
Study Goals	After completing the course Advanced Concept Design the students will be able: Concept Design 1 1.1 to define a design vision based on the insights coming from design related research in the areas of expertise supported by their technological investigation, and to develop it further into ideation; 1.2 to apply and reflect on their project management skills in terms of teamwork, communication, time management/planning, consulting people, presentation and reflection. Concept Design 2 2.1 to develop potential ideas into conceptual design by integrating the knowledge and skills gained in the areas of expertise; 2.2 to critically evaluate the desirability, feasibility and viability of the concept design and to advise the client on the actions that are required within a certain timeframe to further develop the concept into a product (service system); 2.3 to apply and reflect on their project management skills in terms of communication, time management/ planning, consulting people, presentation and reflection. Culture 3.1 to gain awareness about the importance of culture for the development of a vision and of concepts; 3.2 to perform a cultural design study: to analyse cultural processes in the context of a design project;	

	3.3 to evaluate findings from a cultural study and to translate them into opportunities for the development of a design vision and concepts; 3.4 to communicate their findings from the cultural study in a report. This includes the use of visuals and text, use of references, as well as oral discussions in meetings. Ergonomics 4.1 to collect and analyse existing knowledge on ergonomic aspects for the Concept Design challenge; 4.2 to conduct research complementing existing knowledge on ergonomic aspects of the Concept Design challenge; 4.3 to evaluate collected knowledge and insights and translate these into opportunities for the Concept Design challenge; 4.4 to communicate about their findings in writing and verbally. Visualization 5.1 Drawing and visualization part: to deploy IDE-Visualization specific drawing methodology and technique, to apply fluency and (i.e.) vigour in sketching lines and shapes, to visually explore and elaborate when running a design trajectory/ process, and to tactically and effectively weigh the way how to visualize. 5.2 Portfolio part: to create, include, structure and select project representative -graphic- materials (content), to 'compose' a personal portfolio (form), and to clearly describe one's vision and ambitions as a designer (personal).
Education Method	Both, individual and group work are required in the course Advanced Concept Design. The educational format depending on the area of expertise and project consists of: Lectures/ Tutorials Exercises/ Workshops Q&A Sessions / Forum Tutorial Assignment(s) Project Work
Assessment	For the group work of the first quarter, students form teams, guided by a design coach and area specific experts. In the second quarter, students work individually on their concept development, receive supporting knowledge from the areas of expertise and get guidance and feedback from their design coach. During the whole semester, there will be weekly Visualization classes to train students individual communication and presentation skills. The assessment of the course Advanced Concept Design is based on the quality of oral, written and visual (e.g. movie, drawings) deliverables from the different areas of expertise and the Concept Design Project. Students are assessed both as a group and individually. The assessment overview including Fail and Pass regulation can be found in the ACD Manual as published on the Brightspace website. More details on the assignments of the areas of expertise and the project can also be found in the ACD Manual with references to the different (sub)components of the course. The grades for the 5 separate ACD components will be published in Osiris, as well as the ACD total grade if the course has been successfully completed.
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details.
Special Information	Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty. Special Information ACD Course coordinator Dr. Ir. Marijke Dekker Room C-3-170, phone 83028 E-mail: m.c.dekker@tudelft.nl Involved co-coordinators; Dr. ir. Sonja Paus Buzink - Ergonomics Dr. ir. Annemiek van Boeijen - Culture Ir. Jan Willem Hoftijzer - Visualization
Remarks	This is a collaboration course. External partners bring in real life cases and obtain IP rights to the results. At the start of the course the course staff will ask you to sign the IP(R) transfer with TU Delft to make collaboration possible. More detailed information can be found in the student portal -> practical affairs -> Intellectual Property in IDE education -> collaboration courses.

ID4175-23	Advanced Embodiment Design Project	10
Course Coordinator	Dr. J. Wu	
Contact Hours / Week x/x/x/x		
Education Period	3 4	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	The course Advanced Embodiment Design Project is about bringing a product a step further in the design process, with a focus on integrated embodiment design. In a team of 6 students, you are going to work on a client-based project for either a company or a research project and end up with a validated prototype and improved design. This can be a proof of principle design or a later stage design step, to bring a product to the production stage. AEDP WILL TEACH YOU: - To work professionally in a team and deal with the dynamics within a team for a full semester. - To independently lead and work on a design project. - Communicate professionally with both the coach as the client orally and on paper. - Engineer a concept design to a near-production ripe product.	
Study Goals	After completing the course Advanced Embodiment Design Project the students will be able to: 1 Evaluate and analyze an existing product by examining its primary function, subfunctions, and its material composition and manufacturing processes. 2 Apply knowledge in (at least) two of the following expertise domains to optimize the product: physical and cognitive ergonomics, product experience and aesthetics, or circular and environmental impact. 3 Create new embodiment solutions to deliver improved products, via modelling and simulations, physical and virtual prototypes. 4 Evaluate and analyze embodiment solutions by conducting comprehensive test concerning feasibility, viability, and desirability. 5 Communicate design outcomes through product demonstrations, presentations, and a final technical report. 6 Evaluate and reflect on experiences and learnings gained while collaborating within a design team.	
Education Method	Within AEDP you will work together in a team on a client-based embodiment project. All available projects will be presented during the Kick-off meeting of the course, in which approximately 50% of the projects is related to companies and 50% of the projects is related to research projects. During the semester we have setup different Key-note Lectures which consisting of helpful topics for your project: Conceptual and Advanced Prototyping methods, Technology Readiness Levels, Systems Engineering, Design for Advanced Manufacturing (Additive and Subtractive), becoming a Reflective Designer, etc. During the complete semester you will be coached by one of our coaches on a weekly basis, supporting the design process and guiding you as a group to a positive end result.	
Assessment	AEDP 0: RE-WRITTEN ASSIGNMENT - Team deliverable - Formative - Weight: 0% Description: Together with the client and coach the assignment is re-written, to be sure that everyone has the same expectations of the project and the project outcome. Expected result: Please refer to Brightspace for more information. This will be posted a few weeks prior to the assignment. AEDP 1: MID-TERM REPORT - Team deliverable - Formative - Weight: 0% Description: A report that shows all the steps made so far. Expected result: Please refer to Brightspace for more information. This will be posted a few weeks prior to the assignment. AEDP 2: MID-TERM PRESENTATION - Team deliverable - Formative - Weight: 0% Description: The mid-term presentation is a check-up for the client, course coordinators, coaches and fellow students. The presentation will show the defined product, challenges involved, the approach and the concerning planning for the second quarter. It should also show the progress and steps for implementation of all expertise areas. Expected Result: The presentation will take around 12 minutes with 8 minutes of Q&A afterwards. Please refer to Brightspace for more information. AEDP 3: GROUP REFLECTION - Team deliverable - Formative - Weight: 0% Description: A report in which the group identifies lessons learned, where they discuss, prioritize and integrate what has been learned into a strategy for action to improve future performance (Q4). Expected Result: A maximum of 1000 words. Make sure to include relevant pictures and figures of the teamwork. AEDP 4: FINAL REPORT - Team deliverable - Summative	

- Weight: 50% :

Description:

A to the point technical report that shows all the steps made so far.

Expected Result:

Please refer to Brightspace for more information.

AEDP 5: FINAL PRESENTATION

- Team deliverable

- Summative

- Weight: 40%

Description:

The final presentation shows the challenges, achievements reflection and the deliverables. The presentation should also contain a video and the prototype.

Expected Result:

The presentation will take around 12 minutes with 8 minutes of Q&A afterwards. Please refer to Brightspace for more information. 22

AEDP 6: FINAL VIDEO

- Team deliverable

- Formative

- Weight: 0%

Description:

The video is a narrative of the final result of the course and the design process leading to it.

Expected Result:

A maximum 4 min long video of your final result and the design process. The video is delivered together with your final report.

AEDP 7: PERSONAL REFLECTION

- Individual deliverable

- Summative

- Weight: 10%

Description:

A report in which the student identifies lessons learned and discusses, prioritizes and integrates what has been learned into a strategy for action to improve future performance as a professional designer.

Expected Result:

A report of maximum 1000 words. Make sure to include relevant images and figures of the project.

Enrolment / Application

For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose.

Registration periods and deadlines are communicated on the website. See <http://www.io.tudelft.nl/osiris> for exact deadlines and details.

Registration beyond the registration deadlines is not possible.

These rules apply to IDE students as well as to students from outside the IDE faculty.

Remarks

This is a collaboration course. External partners bring in real life cases and obtain IP rights to the results.

At the start of the course the course staff will ask you to sign the IP(R) transfer with TU Delft to make collaboration possible.

More detailed information can be found in the student portal -> practical affairs -> Intellectual Property in IDE education -> collaboration courses.

ID4180	Managing Product Innovation	3
Course Coordinator	Ir. R.G.H. Bluemink	
Contact Hours / Week x/x/x/x		
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The Managing Product Innovation course teaches fundamental and practical aspects of product innovation, focusing on three domains: Structuring the Product Innovation Process, New Product Economics, and Stakeholder Management.	
Study Goals	Upon successful completion of the course, you will be able to: 1) Set up and manage your product-service innovation project to achieve the formulated objectives and milestones 2) Set up a business case for your product-service concept aligned with the stakeholders' interests 3) Communicate effectively with the project stakeholders	
Education Method	The course combines weekly lectures with interactive discussions, small group exercises, group presentations, and peer assessment. Group assignments are used to practice knowledge and the skills conveyed during lectures. Alignment with the ID4170 Advanced Concept Design course (ACD) enables students to work in the same groups in both courses and apply their learnings directly in the ACD group assignment.	
Assessment	The course combines weekly lectures with interactive discussions, small group exercises, group presentations, and peer assessment. Group assignments are used to practice knowledge and the skills conveyed during lectures. Alignment with the ID4170 Advanced Concept Design course (ACD) enables students to work in the same groups in both courses and apply their learnings directly in the ACD group assignment.	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

ID4185-23	Strategic and Sustainable Design	3
Course Coordinator	B. Sprecher	
Contact Hours / Week x/x/x/x		
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	This course focuses on the strategic reasoning of how and why various organisations choose for sustainable innovations. We will be working with case studies from the Harvard Business School (HBS), in order to help you develop problem-solving and decision-making skills in a business context. Students will be discussing the fundamentals of sustainability in business, such as economic impact, company strategy, and operationalisation and will benefit from recent research on this topic conducted at the IDE faculty. The course will touch upon various frameworks for sustainability, such as circular economy, and provides an opportunity to develop academic writing skills and support the Advanced Embodiment Design course.	
Study Goals	After completing the course students will be able to: - perform a case study on a chosen (design) innovation, using theory related to the three pillars of sustainable development (people, planet, and profit) - explain how different drivers influence the sustainability choices made by a company - Evaluate the business case for sustainable innovation in different organizational contexts - communicate about sustainable innovation both in business language and scientific language	
Education Method	The course is structured around discussion and several conference-style presentations, and will have weekly working sessions. Prior to each session students will prepare by reading/viewing part of the course material on Brightspace. Students will also be challenged to find, assess and compare information regarding sustainable innovations in practice.	
Literature and Study Materials	The course will use material made available via Brightspace (e.g. scientific papers, CSR reports, documentaries).	
Assessment	In groups, students will write two well-researched scientific case studies on a real sustainable innovation project, either as a prepared business case, or real-world case study. The first case study report is 20% of the grade, the second report is 80% of the final grade.	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

Year	2023/2024
Organization	Industrial Design Engineering
Education	Master Integrated Product Design

2nd year Master Industrial Product Design

ID4190-16	Graduation Project (IPD)	30
Course Coordinator	Prof. G.W. Kortuem	
Contact Hours / Week x/x/x/x		
Education Period	None (Self Study)	
Start Education	1	
	2	
	3	
	4	
	5	
Exam Period	none	
Course Language	English	
Course Contents	The graduation project gives the student the opportunity to show he is worthy of the Master of Science title and that he fulfils the requirements of the program. This implies that the student uses his knowledge and skills as an independent industrial design engineer to execute a complex project. The graduation project can be seen both as the masterpiece of the student, and as a (personal) learning experience. The scope is therefore not only on testing competences, but also on the development of knowledge, understanding and skills of the student during the project. The graduation project is considered to be a stepping stone to a future professional career. A high level of independence is therefore expected from the student in the planning and execution of the assignment and in the acquisition of knowledge, understanding and skills. A student may start the graduation project under the condition that he has passed all mandatory/first year courses of the Masters Program and the Project Brief is approved by the Board of Examiners.	
Study Goals	On completion of this course, the student will be able to: - effectively collect, analyse, generate and evaluate knowledge required for the project - justify his/her choices with respect to used methods and/or approaches used in the project. - deliver a relevant (feasible, viable and desirable) project result - effectively and thoroughly communicate to- and discuss with stakeholders involved in the project - manage a design/research project independently within the given time	
Assessment	During the graduation project, there are three formal moment when the students work is assessed; Mid-term, Green-light and final assessment. The assessment of the graduation projects is supported by means of a rubric. The deliverables for the graduation project consist of; - A report providing insight in the process and project results, - A showcase; either a prototype or model, a movie or a poster - A public presentation (at the graduation day) An evaluation of the project with the supervising team, consisting of a chairman and a mentor from IDE and, when applicable, an external mentor is scheduled directly following the public presentation.	
Remarks	A more elaborate description of the graduation project can be found in the document Graduation Manual. (See; http://io.studenten.tudelft.nl/ - education - Graduation and Internship - Graduation (MSc)- Download - Graduation manual for students)	

Year	2023/2024
Organization	Industrial Design Engineering
Education	Master Integrated Product Design

Electives (min. 30 EC) IPD

Year	2023/2024
Organization	Industrial Design Engineering
Education	Master Integrated Product Design

Specialization Medesign

Year	2023/2024
Organization	Industrial Design Engineering
Education	Master Integrated Product Design

Compulsory Electives (min. 9 ECTS) Medesign
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ID5301	Biomechanics	3
Course Coordinator	Dr. T. Huysmans	
Contact Hours / Week x/x/x/x		
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Digital human models (DHM) are computer representations of humans that allow to simulate the interaction of the digital human with a virtual product or environment. Since their introduction in the 1960s, enormous progress has been made but it is still a very active research area due to the complex nature of the human body. A variety of DHM software tools are currently available to efficiently incorporate human factors early-on in the design process and they are used to improve function, fit, and comfort of products in e.g. automotive, aviation, defense, fashion, sports, and medical industries. This course focusses on two emerging DHM areas: 3D anthropometry and biomechanical modelling and simulation, with a focus on the musculoskeletal system (MSK). The course will start with an overview of the function of bones, muscle, and tendons of human body and introduce basic biomechanical concepts. In a flipped classroom setting, students will familiarize themselves with the range of available DHM technologies and tools. Through example cases and a group assignment, students will learn to analyze (dynamic) human-product interactions, how to optimize them using DHMs, and finally how to set up a physical validation study to validate design choices. Example cases include optimizing bicycle seat height and improving respiratory mask fit. Typical products that are treated are tools, supports (chairs and beds), shoes, wearables, and medical equipment.	
Study Goals	After completing the Digital Human Modeling course, students should be able to: LO 1: Explain the general biomechanical functions of bone, muscles, ligaments, tendons, and joints of the human body; LO 2: Apply simple biomechanical modelling techniques on paper; LO 3: Classify digital human modeling (DHM) tools according to purpose and underlying technology; LO 4: Analyze human-product interactions and identify critical body dimensions and product parameters that affect the interaction; LO 5: Employ existing 3D anthropometry and musculoskeletal (MSK) simulation software to analyze fit and biomechanics of simple human-product interactions; LO 6: Propose product modifications to improve product performance in terms of function, fit, safety and comfort; LO 7: Identify an approach (hardware, software, criteria) for evaluation of fit and biomechanics in practical human-product interactions.	
Education Method	lectures, flipped classroom, workshop, and peer instruction.	
Literature and Study Materials	Reader, course slides, scientific articles, DHM software.	
Assessment	group assignments	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

ID5383	Inclusive Design	3
Course Coordinator	Dr. S.U. Boess	
Contact Hours / Week x/x/x/x		
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The student has learned the design relevant terms and organisational structures in the world of inclusive design (and how it is different from and similar to design for all, or universal design). The student has practiced inclusive design in a case study (group work) for a specified population as mentioned. The student has involved a few persons of her or his target group in the assignment.	
Study Goals	The main course goal is to understand the current state of the phenomenon of Inclusive Design, which had alternative names like Design for All or Universal Design. Inclusive Design means to aim for the widest possible audience during and with the design process. It does not mean however to design for all 7 billion people on earth! It means considering any specified population to include in designing, and how solutions could also benefit other groups or a wider group. For example: left-handed people, wheelchair users, tall people, small people, obese people, colour blind people, people with dyslexia or RSI, people with age-related specifics etc. The list continues, which means that we realise that hardly anyone is 'normal'. People aged over 65 for example, currently count 15% in the Netherlands but this will grow to 25% in 2050. They are more diverse than younger people. Risks for them also rise: lethal home accidents rise 5-fold compared to younger people. The effect of design should be that participation and quality of life is enhanced, and stigmatisation and risk is avoided. Take the example of eye glasses: they were an orthosis in the middle ages and are now aesthetic lifestyle enhancements. To design inclusively, the designer needs to be able to collaborate with people, understand understand their usage of their everyday things, and respond creatively.	
Education Method	(Guest) lectures in combination with group work on design for and with specified populations.	
Literature and Study Materials	Simon Keates & John Clarkson (2004): Countering design exclusion, an introduction to inclusive design. Springer. Websites: http://www.inclusivedesigntoolkit.com and http://www.hhrc.rca.ac.uk/	
Assessment	The assessment is based on - course participation - team oral presentation and final report - a short written individual essay.	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	s.u.boess@tudelft.nl	

ID5663	Contextmapping Skills	6
Course Coordinator	Prof.dr. P.J. Stappers	
Contact Hours / Week x/x/x/x		
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Participation in 'Contextmapping Skills' is open to all MSc students who believe that end-users are important sources of information for leading new product and service development. Because the elective builds upon the theory and practice taught in ID4216 ('C&C' required for Dfl and SPD), following that course prior to, or in parallel, is highly recommended. Participants are asked to supply a short written motivation and (Manage your Master) portfolio about his or her experience with user research (to both P.J.Stappers@TUDelft.nl and Benjamin@Muzus.nl) to help us prepare the course. Furthermore, we highly recommend you to plan enough time for this course (at least 8 hours for groupwork, including fieldwork, available per week).	
Course Contents	Contextmapping is a design (research) method for including users and other stakeholders in a design process as 'expert of their experience'. By applying generative techniques in the research process, underlying needs, opportunities, motivations and dreams can be discovered. Over the past two decades, the method and its underlying principles and theories have been developed at TU Delft in close collaboration with Dutch and international researchers and practitioners. This development has been done in about a dozen PhD projects, hundreds of MSc projects, and two (online) masterclasses, most of these in close collaboration with industry (see http://studiolab.ide.tudelft.nl/studiolab/contextmapping/resources/). The Master elective course 'Contextmapping Skills' gives students in-depth experience and a chance to develop their skills at these techniques. It is the best opportunity to get specific training from the experts on the method, in a project with a real-world brief and client. The course has a hands-on character, combining elements of reviewing the state of the art, scientific principles, and exploring and practising methods of - determining focus and scope - recruiting participants - designing sensitising materials and workshop materials - setting up, conducting generative group sessions or interviews - analysing field data - creating concept designs grounded in use insights - communicating the insights and designs to stakeholders and clients.	
Study Goals	Upon completion of the course, the student will be able to - understand theories underlying contextmapping techniques - understand, select, and evaluate techniques for a project - apply the techniques in the field - extend the techniques for a specific target group or situation	
Education Method	The course covers both theory and practice. Both are taught in close collaboration between TU staff (http://studiolab.ide.tudelft.nl/studiolab/contextmapping/) and service design agency Muzus (http://www.muzus.nl/en/). The theory part consists of lectures and case examples. In the practical part design research techniques are applied by the students in a project for a 'real' client. In small groups, students will perform the whole contextmapping process in close collaboration with the client, user and tutors. Results will be implemented by the company, helping them towards user-centred innovation.	
Assessment	Assessment is done on the basis of work performed during the course. In part this is judged on the basis of the contribution to the research elements, in part a portfolio / final report and a presentation / workshop at the client is assessed. The precise method of assessment is made known at the first meeting of the group.	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

ID5721	Capita Selecta Medisign	3
Course Coordinator	Dr.ir. M. Melles	
Contact Hours / Week x/x/x/x	-	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The field of design for healthcare is broad and diverse. As a medical design engineer you will work in different positions with professionals from various backgrounds. In this course, the broadness of the field of healthcare design is explored by means of a series of guest lectures of healthcare designers, medical professionals, Medisign alumni, healthcare design researchers and Medisign graduates. Medisign students will become acquainted with different perspectives on design for health and have the opportunity to build their professional network. This course encourages an exchange between professionals and Medisign students during the lectures at which design challenges and research topics in the medical field are presented and discussed. The course will prepare students for their graduation in healthcare; they will learn about the different domains in healthcare, learn how and where design can play a role in healthcare, and learn what their role as designer can be. Following this course is highly recommended for all students following the Medisign specialisation.	
Study Goals	The main objective of this course is to get acquainted with the broad landscape of design for healthcare and to bring students into contact with professionals who are in different ways involved in the development of products or services for healthcare. This way the student is encouraged to reflect on their medical design skills and preferences for a future career. At the end of this course the student should be able to differentiate between the different roles a medical design engineer can have and recognise the different perspectives of healthcare professionals, engineers, and researchers on medical product development.	
Education Method	The course consists of a series of guest lectures for which guests are invited from different areas within the field of medical design (e.g., healthcare designers, medical specialists, Medisign graduates, medical researchers from various backgrounds). Next to this, students need to work on a short design assignment to explore the possibilities for a Medisign graduation project linked to a patient and/or health professional. Guest lecturers are often interested in initiating a graduation project or offer an internship. Capita Selecta encourages and facilitates these opportunities.	
Literature and Study Materials	Study materials will be available via Brightspace.	
Assessment	Students need to attend all 7 lectures, actively take part in the discussions, and carry out a short design assignment. The assessment of this course is formative.	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Dr.ir. Marijke Melles E-mail: m.melles@tudelft.nl	

ID5732	Health Psychology: Tools and Methods	3
Course Coordinator	Dr. J.J. Kraal	
Contact Hours / Week x/x/x/x		
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	When doing design research designers often encounter psychological topics. Especially in healthcare related projects, topics such as motivation during rehabilitation, coping with stressful situations, and medication adherence require understanding of psychological theory and methodology to come to a good design. Yet psychological methods (including questionnaires and patient-reported outcome measures) and models are not commonly embedded in the design process. Using validated questionnaires will help designers to execute high quality research and improve the reliability of their outcomes. Furthermore, knowledge of psychological models will help designers understand which psychological factors influence behavior and will inspire and guide them in their design process. This course provides insight into the content and applicability of psychological models & methods for design research in the healthcare industry. The content will be taught through a series of theoretical lectures and practical assignments that serve as input and deeper thought about the lecture content. Parallel to the lectures, students will work on a group design project under supervision of a design coach.	
Study Goals	LO1: Describe the basic elements (e.g. concepts, variables) in psychological theory. LO2: Explain how individual differences and the social environment influence health using the psychological theories covered in the lectures. LO3: Identify and analyse at least one example of a design that has used psychological theory in the design process (e.g. definition, ideation, testing). LO4: (Re)design a product, service, or product-service system by applying models and theories from health psychology. LO5: Write and justify a research proposal to evaluate a theory-based design aimed at preventing, improving, or maintaining health.	
Education Method	The students follow lectures (2 hours a week). A design project is interwoven throughout the course and used to apply theoretical and methodological insights. Students carry out the design project as a group but also submit an individual reflection. The design project takes approximately 6 hours per student per week.	
Literature and Study Materials	Journal articles and selected book chapters.	
Assessment	The students' work is assessed on the following items: Research proposal Description and explanation of the health phenomenon, analyses of design examples, and specification of methodology Design process: Theoretical argumentation of design choices, consistency of design with context, and credibility of the design. Reflection: Structured critical reflection of the strengths and weaknesses of the selected theory, methodology and design	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

ID5741-23	eHealth Design	3
Course Coordinator	Dr. V.T. Visch	
Contact Hours / Week x/x/x/x		
Education Period	3	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	This master elective course focuses on the developments within the eHealth sector and enables students to gain more knowledge about this growing area for healthcare design. Lectures will be provided by the members of the eHealth research group within the IDE faculty. Course topics run parallel to the eHealth research topics at our faculty including user profiles towards health, health and social economic positions, motivation and health, health journeys, health and game design, health and AI interaction, health feedback design, and remote care. In the course you will analyse and present a concept for an improved redesign for an existing eHealth app. These apps will be provided by eHealth developers who will give guest presentations and feedback to your design during the course.	
Study Goals	LO1: The students are able to explain what eHealth is from an industrial design perspective. LO2: The students are able to critically analyze eHealth products in practice based on scientific knowledge. LO3: The students are able to create a concept for a redesign of an existing eHealth product. LO4: The students are able to recommend their eHealth redesign based on theoretical- and practice-based arguments.	
Education Method	The students follow lectures and work in a team of 2 on an eHealth design assignments throughout the course. Every week a new topic is introduced by research staff and guest lecturers of eHealth developers. Knowledge gained during these different lectures have to be applied while working on the assignment. Each lecture is followed by a coaching session hosted by the lecturer. Sharing and discussing insights and knowledge with fellow students, tutors, and eHealth developers (e.g. gathered in desktop research) take a central place in this course. Intermediate and final results are presented to the stakeholders, tutors and fellow students.	
Literature and Study Materials	Literature will be provided during each lecture. With the weekly announcements of the lectures (1 week in advance, through Brightspace) the student can also be asked to go through 1 or 2 relevant papers before attending the lecture. Literature includes several scientific articles or book chapters related to the four different topics. Additionally, up-to-date websites and examples of e-Health applications are provided.	
Assessment	The students' work is assessed on the following two items: Result: Quality and depth of the content of the presentation, report and redesign, capability of linking lecture knowledge to practice. For the report, a template is provided. Communication / presentation: Quality of execution of presentation, showing interest and motivation. Personal communication ability, presenting convincing arguments for own eHealth design, involvement and attitude during joint group discussion.	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

ID5752	Rules and Regulations for Designing Medical Devices	3
Course Coordinator	Dr.ir. S.N. Paus-Buzink	
Contact Hours / Week x/x/x/x		
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	When it comes to the approval of a new healthcare product onto the market, safety is one of the most important drivers for success. The ecosystem of stakeholders that are responsible for the overall safety regarding a healthcare product is large and complex. Regulatory authorities control the quality and safety of patients and healthcare professionals is safeguarded for newly developed medical devices. This elective provides insights into the medical design directives regarding safety of healthcare products by focusing on the CE marking process (including ISO norms and risk management). The course includes several lectures of TU Delft tutors and external professional(s) working in the healthcare field. Next to the lectures, students will do a duo-assignments in which they apply insights into practice with the use of a case-study. The course shows students the importance of risk analysis of healthcare products and helps them to integrate this in their design process.	
Study Goals	After completing this course you will be able to: LO1: Interpret relevant procedures, rules and regulations on safeguarding quality and safety of products for healthcare. Identify which are applicable Know how to read and apply LO2: Model the ecosystem of stakeholders and their role(s) in the development and approval process of a healthcare product. Identify which are applicable Understand & explain roles LO3: Analyse and manage risks in medical device development. Conduct risk analysis Apply design interventions to minimize residual risks LO4: Communicate in writing and verbally about conformity of the design and design process with European Regulations Compile the required technical documentation Draw up a declaration of conformity	
Education Method	The course includes weekly lectures and workshops, some given by external experts. Next to some self-study of the provided materials you will practice in applying your newly gained knowledge and skills in a duo-assignment. In this duo-assignment, you will work together with one other student on a case study on a medical product-service system. The overall assignment and sub-assignments are interwoven with the lectures and workshops throughout the course. Some examples of sub-assignments are: defining medical device classification, stakeholder mapping, compiling user- and use-environment descriptions, and conducting use-related risk analysis. During the course, you will provide and receive peer feedback on your sub-assignments. Halfway the course there is also an opportunity to receive formative feedback on your work on the case study from the coaches. At the end of the course, you will give a presentation to the group and submit a report about your duo-assignment.	
Literature and Study Materials	Relevant literature will be provided via Brightspace.	
Assessment	During the course you will receive formative feedback on part of your work on the case-study/duo-assignment from peers and one of the coaches. Summative assessment and grading will be based on your participation in the course activities, final report, presentation and communication with the coaches on your case-study/duo-assignment.	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

Ir. R.G.H. Bluemink

Unit	Industrieel Ontwerpen
Department	Methodologie en Organisatie van Design
Telephone	+31 15 27 87273
Room	32.B-4-260

Dr. S.U. Boess

Unit	Industrieel Ontwerpen
Department	Applied Ergonomics and Design
Telephone	+31 15 27 83196
Room	32.B-2-140

Dr. J.H. Boyle

Department	Emerging Materials
-------------------	--------------------

Dr. R.S.K. Chandrasegaran

Department	Methodologie en Organisatie van Design
Room	32.B-4-140

Dr.ir. M.C. Dekker

Unit	Industrieel Ontwerpen
Department	Applied Ergonomics and Design
Telephone	+31 15 27 83028
Room	32.C-3-170

Ir. S.H. Gieles

Department	Design Aesthetics
-------------------	-------------------

Dr. T. Huysmans

Unit	Industrieel Ontwerpen
Department	Applied Ergonomics and Design
Telephone	+31 15 27 81198

Dr. E. Karana

Unit	Industrieel Ontwerpen
Department	Emerging Materials
Telephone	+31 15 27 85726

Prof. G.W. Kortuem

Unit	Industrieel Ontwerpen
Department	Internet of Things
Telephone	+31 15 27 81952
Room	32.B-3-280

Dr. J.J. Kraal

Department	Applied Ergonomics and Design
Telephone	+31 15 27 81747

Ir. A.J.M. van Leeuwen

Department	Human Information Communication Design
-------------------	--

Dr.ir. M. Melles

Unit	Industrieel Ontwerpen
Department	Applied Ergonomics and Design
Telephone	+31 15 27 87514

Dr.ir. S.N. Paus-Buzink

Department	Applied Ergonomics and Design
-------------------	-------------------------------

Dr. rer. nat. C. Schneegass

Department	Human Information Communication Design
-------------------	--

B. Sprecher

Department	Design for Sustainability
-------------------	---------------------------

Prof.dr. P.J. Stappers

Unit	Industrieel Ontwerpen
Department	Design Conceptualization and Communication
Telephone	+31 15 27 85202
Room	32.C-3-130

Dr. V.T. Visch

Unit	Industrieel Ontwerpen
Department	Design Aesthetics
Telephone	+31 15 27 84951
Room	32.C-3-260

Dr. J. Wu

Unit	Industrieel Ontwerpen
Department	Materials and Manufacturing
Telephone	+31 15 27 84858
